CONDITIONAL PROBABILITY DISTRIBUTIONS

Lect. İsmail BAŞOĞLU, Ph.D.

Özyeğin University

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Conditional Probability Distributions

- It is often natural to make probability statements concerning one random variable under the observation of another random variable that is dependent to the first one.
  - X: The odometer value (in km.) of a randomly selected second hand automobile.
  - Y: The second hand price (in ₺) of the same automobile.
  - T: Gender of the last owner of the same automobile (1 if female).
- What is the prob. that a used car is worth less than 60000₺ given that its odometer value is 15000 km?

$$P\{Y < 60000 | X = 15000\}$$

- What is the prob. that a used car is worth more than 100000₺ given that its previous user is a female?

$$P\{Y > 100000 | T = 1\}$$

Conditional Distributions - Discrete Case

- If X and Y are both discrete RVs, it is natural to define the **conditional probability mass function** of X given that $Y = y$, by:
$$p_{X|Y}(x|y) = \frac{P\{X=x,Y=y\}}{P\{Y=y\}} = \frac{p_{XY}(x,y)}{p_Y(y)}$$
under the condition that $P\{Y = y\} > 0$.
- When X and Y are independent, the conditional distribution of X is equal to its marginal distribution.
$$p_{X|Y}(x|y) = \frac{P\{X=x,Y=y\}}{P\{Y=y\}} = \frac{P\{X=x\}P\{Y=y\}}{P\{Y=y\}} = P\{X = x\} = p_X(x)$$

Conditional Distributions - Discrete Case: Example

Q1. Suppose that X and Y have the following joint PMF:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

a. Calculate $P\{X = 2|Y = 2\}$.

b. Find the conditional PMF of X given that $Y = 2$.

c. Calculate $P\{X \geq 2|Y \leq 2\}$.

Conditional Distributions - Discrete Case: Example (cont'd)

A1. The solutions are as follows:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

a. We will expand the conditional probability as always:

$$P\{X=2|Y=2\} = \frac{P\{X=2,Y=2\}}{P\{Y=2\}} = \frac{0.05}{0.2+0.05+0.05} = \frac{1}{6}$$

Conditional Distributions - Discrete Case: Example (cont'd)

A1. The solutions are as follows:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

b. Along with $P\{X=2|Y=2\}$, we also need to compute following probabilities:

$$P\{X=1|Y=2\} = \frac{P\{X=1,Y=2\}}{P\{Y=2\}} = \frac{0.20}{0.2+0.05+0.05} = \frac{2}{3}$$

$$P\{X=3|Y=2\} = \frac{P\{X=3,Y=2\}}{P\{Y=2\}} = \frac{0.05}{0.2+0.05+0.05} = \frac{1}{6}$$

Then we reach to the conditional PMF of $X|Y=2$.

$$p_{X|Y}(x|2) = \begin{cases} 2/3, & x=1 \\ 1/6, & x=2 \\ 1/6, & x=3 \\ 0, & \text{e/w} \end{cases}$$

Conditional Distributions - Discrete Case: Example (cont'd)

A1. The solutions are as follows:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

c. We will expand the conditional probability as always:

$$P\{X \geq 2 | Y \leq 2\} = \frac{P\{X \geq 2, Y \leq 2\}}{P\{Y \leq 2\}} = \frac{0.1+0.1+0.05+0.05}{0.2+0.1+0.1+0.2+0.05+0.05} = \frac{3}{7}$$

Conditional Distributions - Discrete Case: Example II

Q2. Consider the joint PMF of two Bernoulli RVs, X and Y, given in the tabulated format:

$p_{XY}(x,y)$	y	
x	0	1
0	0.40	0.20
1	0.10	0.30

a. Calculate the conditional PMF of X given that $Y = 1$.

b. Calculate $P\{Y = 1|X = 0\}$.

Conditional Distributions - Discrete Case: Ex. II (cont'd)

A2. Answers are as follows:

a. We need to compute the conditional probabilities:

$$P\{X=0|Y=1\} = \frac{P\{X=0,Y=1\}}{P\{Y=1\}} = \frac{0.20}{0.20+0.30} = 0.40$$

$$P\{X=1|Y=1\} = \frac{P\{X=1,Y=1\}}{P\{Y=1\}} = \frac{0.30}{0.20+0.30} = 0.60$$

$$p_{X|Y}\{x|1\} = \begin{cases} 0.40, & x=0 \\ 0.60, & x=1 \\ 0, & \text{e/w} \end{cases}$$

b. $P\{Y=1|X=0\} = \frac{P\{Y=1,X=0\}}{P\{X=0\}} = \frac{0.20}{0.40+0.20} = \frac{1}{3}$

Conditional Distributions - Continuous Case

- If X and Y are continuous RVs, it is natural to define the **conditional probability density function** of X given that $Y = y$, by:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$$

under the condition that $f_Y(y) > 0$.

- Notice that, computing a conditional PDF of X on Y requries computing the **marginal PDF** of Y.
- **When X and Y are independent,** the conditional distribution of X is equal to its marginal distribution.

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_X(x) f_Y(y)}{f_Y(y)} = f_X(x)$$

Conditional Distributions - Continuous Case: Example

Q3. The joint density of X and Y is given by

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0<x<1, 0<y<1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Compute the conditional PDF of X given that $Y = 0.75$.

b. Calculate $P\{0.2 \leq X \leq 0.5 | Y = 0.75\}$.

A3. Both questions are about conditioning X on Y. Therefore, we need to compute the marginal PDF of Y, first.

$$f_Y(y) = \int_0^1 (2-x-y)\,dx = \left[2x - \frac{x^2}{2} - yx\right]_{x=0}^{x=1} = \frac{3}{2} - y, \quad y \in (0,1)$$

Conditional Distributions - Continuous Case: Ex. (cont'd)

a. The conditional PDF of $X|Y = 0.75$ is:

$$f_{X|Y}(x|0.75) = \left.\frac{f_{XY}(x,y)}{f_Y(y)}\right|_{y=0.75} = \left.\frac{2-x-y}{\frac{3}{2}-y}\right|_{y=0.75} = \frac{2-x-0.75}{\frac{3}{2}-0.75} = \frac{5-4x}{3}$$

for $x \in (0,1)$

b. The requested conditional probability can be found using the conditional PDF of $X|Y = 0.75$:

$$\begin{aligned} P\{0.2 \leq X \leq 0.5|Y = 0.75\} &= \int_{0.2}^{0.5} f_{X|Y}(x|0.75)\,dx \\ &= \int_{0.2}^{0.5} \frac{5-4x}{3}dx \\ &= \left[\frac{5x-2x^2}{3}\right]_{0.2}^{0.5} = \frac{9}{25} \end{aligned}$$

Conditional Distributions - Continuous Case: Example II

Q4. The joint density of X and Y is given by

$$f_{XY}(x,y) = \begin{cases} \frac{12}{5}x(2-x-y), & 0<x<1, 0<y<1 \\ 0, & \text{otherwise} \end{cases}$$

a. Compute the conditional PDF of X given that $Y=y$.

b. Compute the conditional PDF of X given that $Y=0.5$.

c. Calculate $P\{0.2 \leq X \leq 0.8 | Y=0.5\}$.

A4. All three questions are about conditioning X on Y. Therefore, we need to compute the marginal PDF of Y, first.

$$\begin{aligned} f_Y(y) &= \int_0^1 \frac{12}{5}x(2-x-y)\,dx = \int_0^1 \frac{24x-12x^2-12xy}{5}dx \\ &= \left[\frac{12x^2-4x^3-6x^2y}{5}\right]_{x=0}^{x=1} = \frac{8-6y}{5}, \quad y \in (0,1) \end{aligned}$$

Conditional Distributions - Continuous Case: Ex. II (cont'd)

a. The conditional PDF of $X|Y=y$ is:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{\frac{12}{5}x(2-x-y)}{\frac{8-6y}{5}} = \frac{6x(2-x-y)}{4-3y}, \quad x \in (0,1)$$

b. The conditional PDF of $X|Y=0.5$ is:

$$f_{X|Y}(x|0.5) = \left.\frac{6x(2-x-y)}{4-3y}\right|_{y=0.5} = \frac{18x-12x^2}{5}, \quad x \in (0,1)$$

c. The requested conditional probability can be found using the conditional PDF of $X|Y=0.5$:

$$\begin{aligned} P\{0.2 \le X \le 0.8 | Y=0.5\} &= \int_{0.2}^{0.8} f_{X|Y}(x|0.5)\,dx \\ &= \int_{0.2}^{0.8} \frac{18x-12x^2}{5}\,dx \\ &= \left[\frac{9x^2-4x^3}{5}\right]_{0.2}^{0.8} = 0.6768 \end{aligned}$$

Conditional Distributions - Continuous Case: Example II

Q5. The joint PDF of X and Y is given by

$$f_{XY}(x,y) = \begin{cases} \frac{1}{y}e^{-\left(y+\frac{x}{y}\right)}, & 0 < x < \infty, 0 < y < \infty \\ 0, & \text{elsewhere} \end{cases}$$

a. Compute the conditional PDF of X given that $Y = y$.

b. Compute the conditional PDF of X given that $Y = 1$.

c. Calculate $P\{1 \leq X \leq 2 | Y = 1\}$.

A5. All three questions are about conditioning X on Y. Therefore, we need to compute the marginal PDF of Y, first.

$$\begin{aligned} f_Y(y) &= \int_0^\infty \frac{1}{y}e^{-\left(y+\frac{x}{y}\right)}dx = \left[-e^{-\left(y+\frac{x}{y}\right)}\right]_{x=0}^{x=\infty} \\ &= 0 - \left(-e^{-y}\right) = e^{-y}, \quad y \in (0,\infty) \end{aligned}$$

Conditional Distributions - Continuous Case: Ex. III (cont'd)

a. The conditional PDF of $X|Y=y$ is:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{\frac{1}{y}e^{-\left(y+\frac{x}{y}\right)}}{e^{-y}} = \frac{1}{y}e^{-\frac{x}{y}}, \quad x \in (0,\infty)$$

b. The conditional PDF of $X|Y=1$ is:

$$f_{X|Y}(x|1) = \frac{1}{y}e^{-\frac{x}{y}}\Big|_{y=1} = e^{-x}, \quad x \in (0,\infty)$$

c. The requested conditional probability can be found using the conditional PDF of $X|Y=1$:

$$\begin{aligned} P\{1 \le X \le 2|Y=1\} &= \int_1^2 f_{X|Y}(x|1)\,dx \\ &= \int_1^2 e^{-x}dx \\ &= [-e^{-x}]_1^2 \approx 0.2325 \end{aligned}$$

Conditional Expectation

- Let $E[X|Y=y]$ denote the conditional expectation of X given that $Y=y$.
  - Note that, the resulting expectation depends on the choice of y, hence, is a function of y.
  - When y is not specified, $E[X|Y]$ itself is a random variable, since it is a function of Y.
- For discrete X and Y:

$$E[X|Y=y] = \sum_{x:p_{X|Y}(x|y)>0} x p_{X|Y}(x|y)$$

- For continuous X and Y:

$$E[X|Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx$$

Computing Expectations by Conditioning

- An immediate use of conditional expectation is that it helps to compute expectations of RVs by conditioning on another RV:

$$E[X] = E[E[X|Y]]$$

- The inner expectation belongs to random variable X, depending on the random choice of Y.
  - Hence, $E[X|Y]$ is a function of Y.
- The outer expectation belongs to random variable Y.
  - It calculates the expected value of a function of Y.

Computing Expectations by Conditioning (cont'd)

- **Theorem:** $E[X] = E[E[X|Y]]$
- **Proof:** We will show it only for the continuous case.

$$
\begin{aligned}
E[E[X|Y]] &= \int_{-\infty}^{\infty} E[X|y] f_Y(y)\, dy \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx \right) f_Y(y)\, dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X|Y}(x|y) f_Y(y)\, dx dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,y)\, dx dy \\
&= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x,y)\, dy \right) dx \\
&= \int_{-\infty}^{\infty} x f_X(x)\, dx = E[X]
\end{aligned}
$$

Computing Expectations by Conditioning: Example

Q6. In order to perform a certain task, Pam invites two of her friends (James and Dwight) who are, independently, likely to respond to this invitation with probability 0.5. The hourly completion time of the task is exponentially distributed with rate parameter $\lambda = Y$ where Y is the number of people (including Pam) who is there to perform the task. Compute the expected time to complete the task.

Computing Expectations by Conditioning: Example

Q6. In order to perform a certain task, Pam invites two of her friends (James and Dwight) who are, independently, likely to respond to this invitation with probability 0.5. The hourly completion time of the task is exponentially distributed with rate parameter $\lambda = Y$ where Y is the number of people (including Pam) who is there to perform the task. Compute the expected time to complete the task.

A6. First, let us determine the PMF of Y.

$$p_Y(y) = \begin{cases} 0.25, & y = 1 \quad (\text{noone responds}) \\ 0.50, & y = 2 \quad (\text{one friend responds}) \\ 0.25, & y = 3 \quad (\text{both friends respond}) \end{cases}$$

Computing Expectations by Conditioning: Example (cont'd)

A6. Let X denote the time to complete the task. We are interested in:

$$E[X] = E[E[X|Y]]$$

We know that:

$$X|Y \sim Exp(\lambda = Y) \Rightarrow E[X|Y] = \frac{1}{Y}$$

Hence:

$$\begin{aligned} E[X] &= E[E[X|Y]] = E\left[\frac{1}{Y}\right] \\ &= p_Y(1) \times \frac{1}{1} + p_Y(2) \times \frac{1}{2} + p_Y(3) \times \frac{1}{3} \\ &= 0.25 \times 1 + 0.50 \times \frac{1}{2} + 0.25 \times \frac{1}{3} \\ &\approx 0.5833 \text{ hours} \end{aligned}$$

Computing Expectations by Conditioning: Example II

Q7. A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to the mine after five hours. Assuming that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until the miner reaches safety?

Computing Expectations by Conditioning: Ex. II (cont'd)

A7. Let X denote the time until miner reaches safety and Y denote the door that he chooses. We will use the conditional expectation formula $E[X] = E[E[X|Y]]$. Note that:

$$\begin{aligned} P\{Y=1\} &= \tfrac{1}{3}, \quad E[X|Y=1] = 2 \\ P\{Y=2\} &= \tfrac{1}{3}, \quad E[X|Y=2] = 3 + E[X] \\ P\{Y=3\} &= \tfrac{1}{3}, \quad E[X|Y=3] = 5 + E[X] \end{aligned}$$

The expected value of X can be found as:

$$\begin{aligned} E[X] &= E[E[X|Y]] \\ &= P\{Y=1\}E[X|Y=1] + P\{Y=2\}E[X|Y=2] + P\{Y=3\}E[X|Y=3] \\ &= \tfrac{1}{3}(2+3+E[X]+5+E[X]) = \tfrac{10}{3} + \tfrac{2}{3}E[X] \end{aligned}$$

The resulting equation helps to solve for $E[X]$:

$$E[X] = \frac{10}{3} + \frac{2}{3}E[X] \Rightarrow E[X] = 10$$

Conditional Variance

- Let $Var[X|Y=y]$ denote the conditional variance of X given that $Y=y$.
  - Note that, the resulting variance is a function of y.
  - When y is not specified, $Var[X|Y]$ itself is a random variable, since it is a function of Y.
- For any two random variables, X and Y:

$$Var[X|Y=y] = E\left[X^2|Y=y\right] - E[X|Y=y]^2$$

Computing Variances by Conditioning

- An immediate use of conditional variance is that it helps to compute variances of RVs by conditioning on another RV:

$$Var[X] = E[Var[X|Y]] + Var[E[X|Y]]$$

- The inner expectation and the inner variance belongs to random variable X, depending on the random choice of Y.
  - Hence, $E[X|Y]$ and $Var[X|Y]$ are functions of Y.
- The outer expectation and the outer variance belongs to random variable Y.
  - They calculate the expected value and the variance of respective functions of Y.

Computing Variances by Conditioning (cont'd)

- **Theorem:** $Var[X] = E[Var[X|Y]] + Var[E[X|Y]]$
- **Proof:**

$$
\begin{aligned}
E[Var[X|Y]] &= E\left[E\left[X^2|Y\right] - E[X|Y]^2\right] \\
&= E\left[E\left[X^2|Y\right]\right] - E\left[E[X|Y]^2\right] \\
&= E\left[X^2\right] - E\left[E[X|Y]^2\right] \\
Var[E[X|Y]] &= E\left[E[X|Y]^2\right] - E[E[X|Y]]^2 \\
&= E\left[E[X|Y]^2\right] - E[X]^2 \\
E[Var[X|Y]] + Var[E[X|Y]] &= E\left[X^2\right] - E\left[E[X|Y]^2\right] + E\left[E[X|Y]^2\right] - E[X]^2 \\
&= E\left[X^2\right] - E[X]^2 = Var[X]
\end{aligned}
$$

Computing Variances by Conditioning: Example

Q8. Consider the task mentioned in Question 6. Compute the variance of the completion time of the task.

Computing Variances by Conditioning: Example

Q8. Consider the task mentioned in Question 6. Compute the variance of the completion time of the task.

A8. First, let us remind ourselves the PMF of Y.

$$p_Y(y) = \begin{cases} 0.25, & y = 1 \quad (\text{noone responds}) \\ 0.50, & y = 2 \quad (\text{one friend responds}) \\ 0.25, & y = 3 \quad (\text{both friends respond}) \end{cases}$$

Let, again X, denote the completion time of the task. We will use the conditional variance formula:

$$Var[X] = \underbrace{E[Var[X|Y]]}_{\text{Step I}} + \underbrace{Var[E[X|Y]]}_{\text{Step II}}$$

We know that:

$$X|Y \sim Exp(\lambda = Y) \Rightarrow E[X|Y] = \frac{1}{Y}, \quad Var[X|Y] = \frac{1}{Y^2}$$

Computing Variances by Conditioning: Example (cont'd)

A8. $Var[X] = \underbrace{E[Var[X|Y]]}_{\text{Step I}} + \underbrace{Var[E[X|Y]]}_{\text{Step II}}$

Step I:

$$Var[X|Y=1] = \frac{1}{1^2} = 1$$
$$Var[X|Y=2] = \frac{1}{2^2} = \frac{1}{4}$$
$$Var[X|Y=3] = \frac{1}{3^2} = \frac{1}{9}$$

$$\begin{aligned} E[Var[X|Y]] &= P\{Y=1\}Var[X|Y=1] \\ &+ P\{Y=2\}Var[X|Y=2] \\ &+ P\{Y=3\}Var[X|Y=3] \\ &= 0.25 \times 1 + 0.50 \times \frac{1}{4} + 0.25 \times \frac{1}{9} \approx 0.4028 \end{aligned}$$

Computing Variances by Conditioning: Example (cont'd)

A8. **Step II:** $Var\left[E\left[X|Y\right]\right] = E\left[E[X|Y]^2\right] - \underbrace{E[E\left[X|Y\right]]^2}_{E[X]^2 \approx 0.3403}$

$$E[X|Y=1]^2 = \left(\tfrac{1}{1}\right)^2 = 1$$

$$E[X|Y=2]^2 = \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}$$

$$E[X|Y=3]^2 = \left(\tfrac{1}{3}\right)^2 = \tfrac{1}{9}$$

$$\begin{aligned} E\left[E[X|Y]^2\right] &= P\{Y=1\}E[X|Y=1]^2 \\ &+ P\{Y=2\}E[X|Y=2]^2 \\ &+ P\{Y=3\}E[X|Y=3]^2 \\ &= 0.25 \times 1 + 0.50 \times \tfrac{1}{4} + 0.25 \times \tfrac{1}{9} \approx 0.4028 \end{aligned}$$

$$Var\left[E\left[X|Y\right]\right] = E\left[E[X|Y]^2\right] - E[E\left[X|Y\right]]^2 \approx 0.2431$$

Computing Variances by Conditioning: Example (cont'd)

A8. **Last Step:**

$$\begin{aligned} Var[X] &= E[Var[X|Y]] + Var[E[X|Y]] \\ &\approx 0.4028 + 0.2431 = 0.6459 \end{aligned}$$

Computing Variances by Conditioning: Example II

Q9. Consider our poor miner in Question 7. What is the variance of the length of time until the miner reaches safety?

A9. Let, again X, denote the time until miner reaches safety and Y denote the door that he chooses. We will use the conditional variance formula:

$$Var[X] = \underbrace{E[Var[X|Y]]}_{\text{Step I}} + \underbrace{Var[E[X|Y]]}_{\text{Step II}}$$

Computing Variances by Conditioning: Example II (cont'd)

A9. $Var[X] = \underbrace{E[Var[X|Y]]}_{\text{Step I}} + \underbrace{Var[E[X|Y]]}_{\text{Step II}}$

Step I:

$$Var[X|Y=1] = 0$$

$$Var[X|Y=2] = Var[3+X] = Var[X]$$

$$Var[X|Y=3] = Var[5+X] = Var[X]$$

$$\begin{aligned} E[Var[X|Y]] &= P\{Y=1\}Var[X|Y=1] \\ &+ P\{Y=2\}Var[X|Y=2] \\ &+ P\{Y=3\}Var[X|Y=3] \\ &= \tfrac{1}{3}(0+Var[X]+Var[X]) = \tfrac{2}{3}Var[X] \end{aligned}$$

Computing Variances by Conditioning: Example II (cont'd)

A9. **Step II:** $Var\left[E\left[X|Y\right]\right] = E\left[E[X|Y]^2\right] - \underbrace{E[E\left[X|Y\right]]^2}_{E[X]^2=100}$

$$E[X|Y=1]^2 = 2^2 = 4$$
$$E[X|Y=2]^2 = (3+E\left[X\right])^2 = 169$$
$$E[X|Y=3]^2 = (5+E\left[X\right])^2 = 225$$

$$\begin{aligned} E\left[E[X|Y]^2\right] &= P\{Y=1\}E[X|Y=1]^2 \\ &+ P\{Y=2\}E[X|Y=2]^2 \\ &+ P\{Y=3\}E[X|Y=3]^2 = \tfrac{1}{3}\left[4+169+225\right] = \tfrac{398}{3} \end{aligned}$$

$$Var\left[E\left[X|Y\right]\right] = E\left[E[X|Y]^2\right] - E[E\left[X|Y\right]]^2 = \tfrac{98}{3}$$

Computing Variances by Conditioning: Example II (cont'd)

A9. **Last Step:**

$$Var[X] = E[Var[X|Y]] + Var[E[X|Y]] = \frac{2}{3}Var[X] + \frac{98}{3}$$

The resulting equation helps to solve for $Var[X]$:

$$Var[X] = \frac{2}{3}Var[X] + \frac{98}{3} \Rightarrow Var[X] = 98$$

Order Statistics

- Suppose we have a collection of n independent observations.

$$x_1, x_2, \ldots, x_n$$

- Suppose further that we have observed the above random variables and their realizations is given in an increasing order.

$$x_{(1)} \leq x_{(2)} \leq \ldots \leq x_{(n)}$$

$$\begin{aligned} x_{(1)} &= \min\{x_1, x_2, \ldots, x_n\} \\ x_{(2)} &= \text{second minimum of } \{x_1, x_2, \ldots, x_n\} \\ \vdots \quad & \quad \vdots \\ x_{(n)} &= \max\{x_1, x_2, \ldots, x_n\} \end{aligned}$$

- $x_{(1)}, x_{(2)}, \ldots, x_{(n)}$ is called the **order statistics** of $x_1, x_2, \ldots, x_n$.

Order Statistics

- We may be interested in the probability distribution of the order statistics.
  - Let $X_1, \ldots, X_n$ be the final exam grades of students in a college course.
  - What is the probability that the student with the maximum grade gets a grade greater than 90?
$$P\{\max\{X_1, X_2, \ldots, X_n\} > 90\} = P\left\{X_{(n)} > 90\right\}$$
  - What is the probability that the student with the minimum grade gets a grade less than 20?
$$P\{\min\{X_1, X_2, \ldots, X_n\} < 20\} = P\left\{X_{(1)} < 20\right\}$$
- How can we identify the distribution of the minimum and the maximum of independent random variables?

Distribution of the Maximum of Two Independent RVs

- Let X_1 and X_2 be two independent RVs with respective CDFs:

$$F_{X_1}(x) = P\{X_1 \leq x\}, \quad F_{X_2}(x) = P\{X_2 \leq x\}$$

- Let $Y = \max\{X_1, X_2\}$ be defined as the maximum of these two random variables.
- Since X_1 and X_2 are both random, it can be $Y = X_1$ as well as it can be $Y = X_2$. Thus, Y is a random variable as well.
- Suppose we are interested in the probability distribution of Y. Let us compute the CDF of Y.

$$\begin{aligned} F_Y(x) &= P\{Y \leq x\} \\ &= P\{\max\{X_1, X_2\} \leq x\} \\ &= P\{X_1 \leq x, X_2 \leq x\} \\ &= P\{X_1 \leq x\} P\{X_2 \leq x\} \\ &= F_{X_1}(x) F_{X_2}(x) \end{aligned}$$

Distribution of the Maximum of n Independent RVs

- Let $X_1, X_2, \ldots, X_n$ be n independent RVs with respective CDFs:

$$F_{X_1}(x) = P\{X_1 \leq x\}, \quad F_{X_2}(x) = P\{X_2 \leq x\}, \quad \ldots \quad , F_{X_n}(x) = P\{X_n \leq x\}$$

- Let $Y = \max\{X_1, \ldots, X_n\}$ be defined as the maximum of these random variables.
- Suppose we are interested in the probability distribution of Y. Let us compute the CDF of Y.

$$\begin{aligned} F_Y(x) &= P\{Y \leq x\} \\ &= P\{\max\{X_1, X_2, \ldots, X_n\} \leq x\} \\ &= P\{X_1 \leq x, X_2 \leq x, \ldots, X_n \leq x\} \\ &= P\{X_1 \leq x\} P\{X_2 \leq x\} \ldots P\{X_n \leq x\} \\ &= F_{X_1}(x) F_{X_2}(x) \ldots F_{X_n}(x) \end{aligned}$$

Distribution of the Maximum of Independent RVs: Ex.

Q10. Suppose that there are 2 batteries, each of which has a lifetime that is independently and exponentially distributed with mean 3 and 5 days respectively. We plug these batteries in a device. It is known that the device stops working if both batteries complete their lifetime. What is the probability that the device will stop working in less than 2 days?

A10. Let:

- X_1 denote the lifetime of the first battery $X_1 \sim Exp\left(\lambda = \frac{1}{3}\right)$
- X_2 denote the lifetime of the second battery $X_2 \sim Exp\left(\lambda = \frac{1}{5}\right)$
- Y denote the lifetime of the device, $Y = \max\{X_1, X_2\}$

The question asks for $P\{Y \leq 2\}$.

Dist. of the Maximum of Independent RVs: Ex. (cont'd)

A10. The question asks for $P\{Y \leq 2\}$.

$$\begin{aligned} P\{Y \leq 2\} &= P\{\max\{X_1, X_2\} \leq 2\} \\ &= P\{X_1 \leq 2, X_2 \leq 2\} \\ &= P\{X_1 \leq 2\} P\{X_2 \leq 2\} \\ &= \left(1 - e^{-\frac{2}{3}}\right)\left(1 - e^{-\frac{2}{5}}\right) \\ &\approx 0.4866 \times 0.3297 \\ &\approx 0.1604 \end{aligned}$$

Distribution of the Minimum of Two Independent RVs

- Let X_1 and X_2 be two independent RVs with respective CDFs:

$$F_{X_1}(x) = P\{X_1 \leq x\}, \quad F_{X_2}(x) = P\{X_2 \leq x\}$$

- Let $T = \min\{X_1, X_2\}$ be the minimum of these two RVs.
- Since X_1 and X_2 are both random, it can be $T = X_1$ as well as it can be $T = X_2$. Thus, T is a random variable as well.
- Suppose we are interested in the probability distribution of T. Let us compute the CDF of T.

$$\begin{aligned} F_T(x) &= P\{T \leq x\} \\ &= P\{\min\{X_1, X_2\} \leq x\} \\ &= 1 - P\{\min\{X_1, X_2\} > x\} \\ &= 1 - P\{X_1 > x, X_2 > x\} \\ &= 1 - P\{X_1 > x\} P\{X_2 > x\} \\ &= 1 - [1 - F_{X_1}(x)][1 - F_{X_2}(x)] \end{aligned}$$

Distribution of the Minimum of n Independent RVs

- Let $X_1, X_2, \ldots, X_n$ be n independent RVs with respective CDFs:

$$F_{X_1}(x) = P\{X_1 \leq x\}, \quad F_{X_2}(x) = P\{X_2 \leq x\}, \quad \ldots \quad , F_{X_n}(x) = P\{X_n \leq x\}$$

- Let $T = \min\{X_1, \ldots, X_n\}$ be defined as the minimum of these random variables.
- Suppose we are interested in the probability distribution of T. Let us compute the CDF of T.

$$\begin{aligned} F_T(x) &= P\{T \leq x\} \\ &= P\{\min\{X_1, X_2, \ldots, X_n\} \leq x\} \\ &= 1 - P\{\min\{X_1, X_2, \ldots, X_n\} > x\} \\ &= 1 - P\{X_1 > x, X_2 > x, \ldots, X_n > x\} \\ &= 1 - P\{X_1 > x\} P\{X_2 > x\} \ldots P\{X_n > x\} \\ &= 1 - [1 - F_{X_1}(x)][1 - F_{X_2}(x)] \ldots [1 - F_{X_n}(x)] \end{aligned}$$

Distribution of the Minimum of Independent RVs: Ex.

Q11. Consider the two batteries with independent lifetimes in Question 10:

$$X_1 \sim Exp\left(\lambda = \tfrac{1}{3}\right), \quad X_2 \sim Exp\left(\lambda = \tfrac{1}{5}\right)$$

Suppose now, the device stops working if one of the batteries completes its lifetime. What is the probability that the device will stop working in less than 2 days?

A11. Let T denote the lifetime of the device, $T = \min\{X_1, X_2\}$. The question asks for $P\{T \leq 2\}$.

However it is more convenient to find $P\{T > 2\}$ since for the minimum of X_1 and X_2 to be greater than 2, they both need to be greater than 2.

Dist. of the Minimum of Independent RVs: Ex. (cont'd)

A11. Let us find $P\{T > 2\}$ first.

$$\begin{aligned} P\{T > 2\} &= P\{\min\{X_1, X_2\} > 2\} \\ &= P\{X_1 > 2, X_2 > 2\} \\ &= P\{X_1 > 2\} P\{X_2 > 2\} \\ &= \left(e^{-\frac{2}{3}}\right)\left(e^{-\frac{2}{5}}\right) \\ &\approx 0.5134 \times 0.6703 \\ &\approx 0.3441 \end{aligned}$$

The probability that the device will stop working in less than 2 days is, then:

$$P\{T \leq 2\} = 1 - P\{T > 2\} \approx 1 - 0.3441 = 0.6559$$

END OF LECTURE SLIDE 09